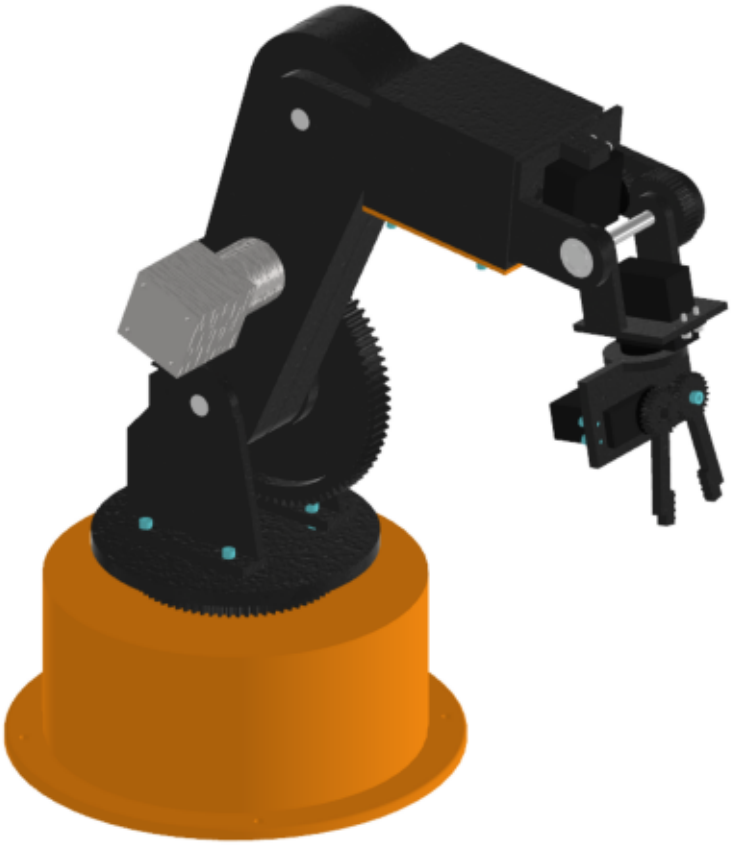
RoboticArm6DOF - calculation report

6-DOF 3D-printed robotic arm, 500 g payload. Jay Patel, Stevens Institute of Technology (ME).

All results are calculated or simulated from the CAD model; no hardware has been built or tested.



Reach (shoulder axis -> TCP, horizontal): 538 mm; L1/L2/L3/D6 = 204/200/192/145.8 mm

Mass (Inventor): moving 1.90 kg, total incl. base 5.02 kg

Assembly: 99 occurrences, 193 constraints, all 0 DOF; FK Inventor vs Python max error 0.0004 mm

J2 shoulder: requirement 6.54 N.m vs capacity 13.30 N.m (SF 2.03)

J3 elbow: requirement 2.81 N.m vs capacity 8.55 N.m (SF 3.04)

FEA A (caseA_upper_arm): 4.88 MPa max, SF 9.2 vs 45 MPa

FEA B (caseB_shoulder_cheek): 12.95 MPa max, SF 3.5 vs 45 MPa

FEA C (caseC_forearm): 9.08 MPa max, SF 5.0 vs 45 MPa

1 Design, actuators and joint torque

Requirement: a six-degree-of-freedom arm that lifts a 500 g payload, modelled in Autodesk Inventor with commercial motors, real bearings and shafts, designed gear reductions, motion simulation, FEA and print-ready parts.

Architecture: J1 base yaw on a 72T printed ring gear (two 6004 bearings), J2 shoulder pitch through an 84T gear bolted to the upper arm, J3 elbow driven by an HTD 5M belt inside the box-section upper arm (motor 90 mm from the shoulder instead of at the elbow, to lower the J2 torque), J4 forearm roll on a 6806 thin-section bearing, J5 wrist pitch on an 8 mm shaft in two 608 bearings, J6 tool roll, and a two-finger gear gripper. J4, J5 and J6 axes meet at the wrist centre (spherical wrist), so position and orientation decouple.

Actuators (commercial): J1 StepperOnline 17HS19-1684S-PG5, J2 and J3 17HS19-1684S-PG19 (NEMA17 48 mm, 1.68 A, 52 N.cm holding; planetary 5.18:1 / 19.2:1, 3 N.m max permissible, 5 N.m moment), J4 and J5 DSSERVO DS3235 (32 kg.cm at 6 V), J6 and gripper DSSERVO DS3218 (about 20 kg.cm at 6 V).

Gear ratios (designed): J1 18:72 m1.5 spur (4:1, total 20.7:1); J2 18:84 m1.75 spur (4.67:1, total 89.6:1); J3 HTD 5M 15:45 belt (3:1, total 57.6:1, stock 375-5M-15 belt, 109.9 mm centres, 4 mm tension slots); J4 20:30 m1 (1.5:1); J5 18:36 m1 (2:1); J6 direct; gripper 24:24 m1.

Joint torque capacity is the smaller of the motor pull-out torque at speed multiplied through the stages, and the gearbox rating (3 N.m) times the external stage. For servos it is 50% of stall times the stage. J2 is sized by the gearbox limit; the 84T output gear was chosen so the gearbox runs at no more than half its rating.

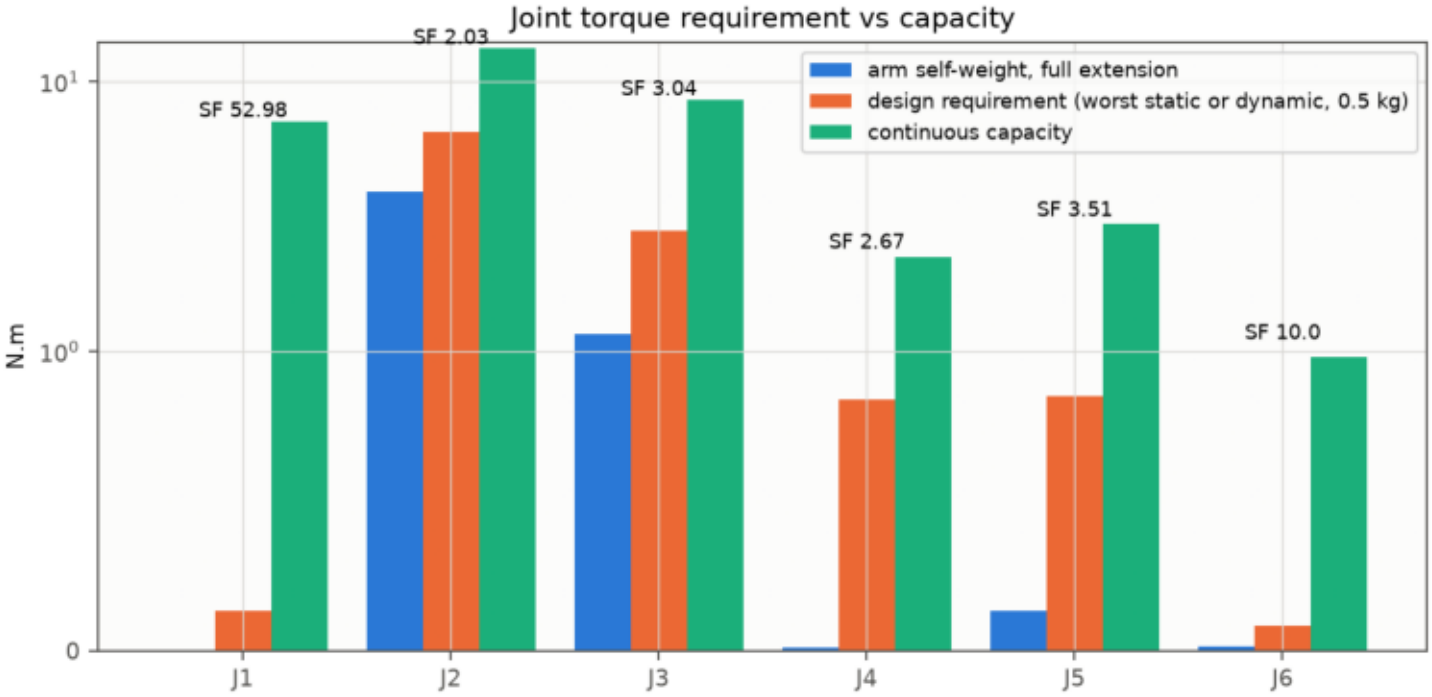
Joint torque table (N.m, worst static = grid search over the measured joint ranges with 0.5 kg payload, trajectory peak = Newton-Euler along the pick-and-place cycle):

Joint	Drive	Ratio	Req static	Traj peak	Design req	Capacity	SF	Speed peak
J1	StepperOnline 17HS19-1684S-PG5 + spur 18	20.72	0.00	0.13	0.13	7.09	52.98	42 d/s
J2	StepperOnline 17HS19-1684S-PG19 + spur 1	89.6	6.54	5.16	6.54	13.30	2.03	26 d/s
J3	StepperOnline 17HS19-1684S-PG19 + HTD5M	57.6	2.81	1.87	2.81	8.55	3.04	14 d/s
J4	DSSERVO DS3235 + spur 20T:30T m1	1.5	0.84	0.02	0.84	2.24	2.67	0 d/s
J5	DSSERVO DS3235 + spur 18T:36T m1	2.0	0.85	0.03	0.85	2.98	3.51	29 d/s
J6	DSSERVO DS3218 + direct	1.0	0.08	0.00	0.08	0.98	10.0	77 d/s

Steppers along the trajectory: J1 motor SF 55.6 (peak 143 rpm), gearbox output peak 0.04 N.m (SF 85.24 vs 3 N.m), max joint speed at 600 rpm 174 deg/s; J2 motor SF 4.39 (peak 383 rpm), gearbox output peak 1.16 N.m (SF 2.58 vs 3 N.m), max joint speed at 600 rpm 40 deg/s; J3 motor SF 10.29 (peak 135 rpm), gearbox output peak 0.66 N.m (SF 4.57 vs 3 N.m), max joint speed at 600 rpm 62 deg/s

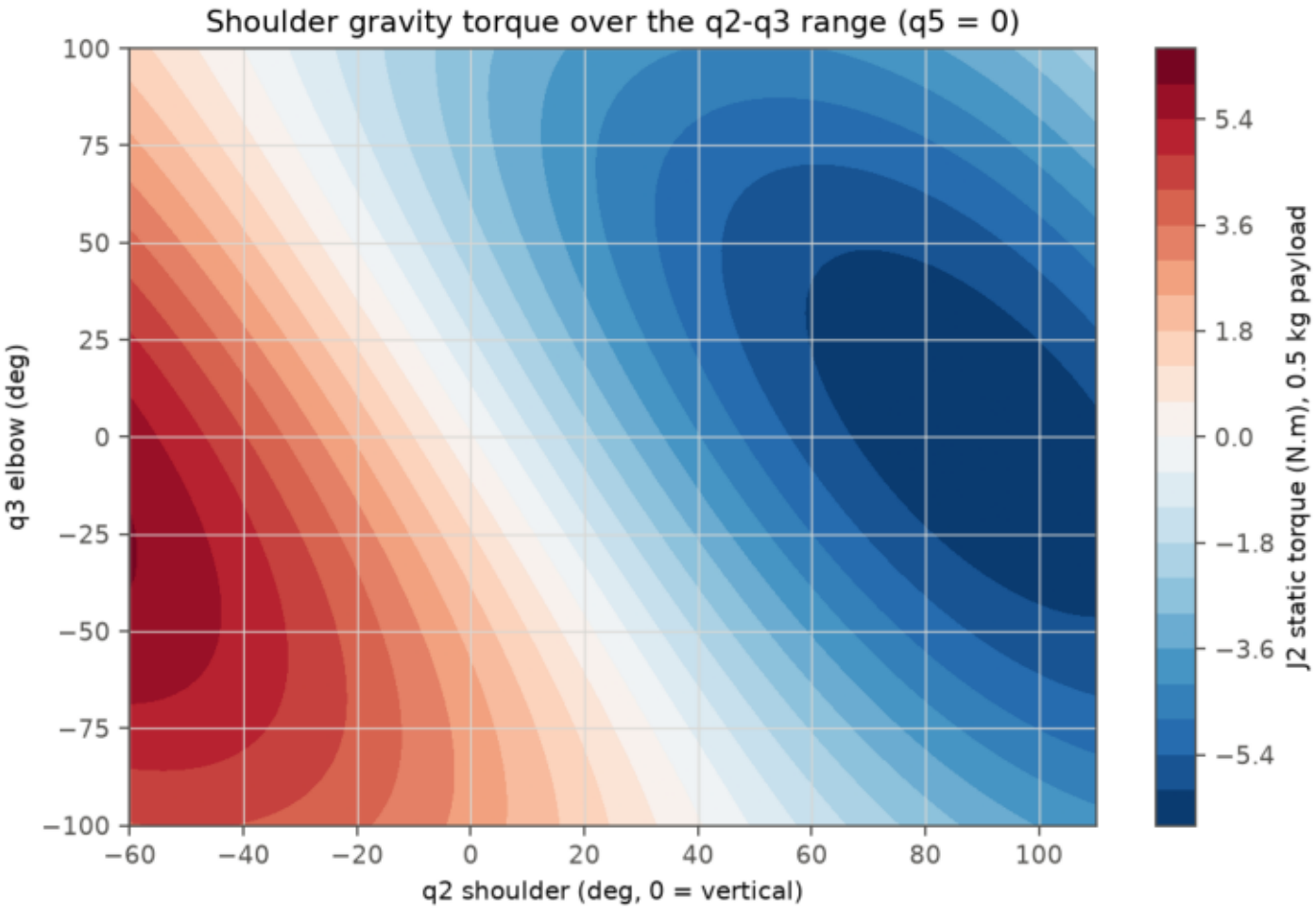
Gripper: needs 5.1 N per jaw (mu 0.5, peak lift acceleration 0.40 m/s2); DS3218 at 50% stall gives 9.6 N at the 51 mm pad lever (SF 1.88).

Joint torque requirement vs capacity



Self-weight at full extension, design requirement (larger of the worst static case and the trajectory peak) and continuous capacity per joint (log scale above 1 N.m).

Shoulder gravity-torque map



J2 static torque over the q2-q3 range with the payload; the maximum is at full horizontal extension.

2 Kinematics and workspace

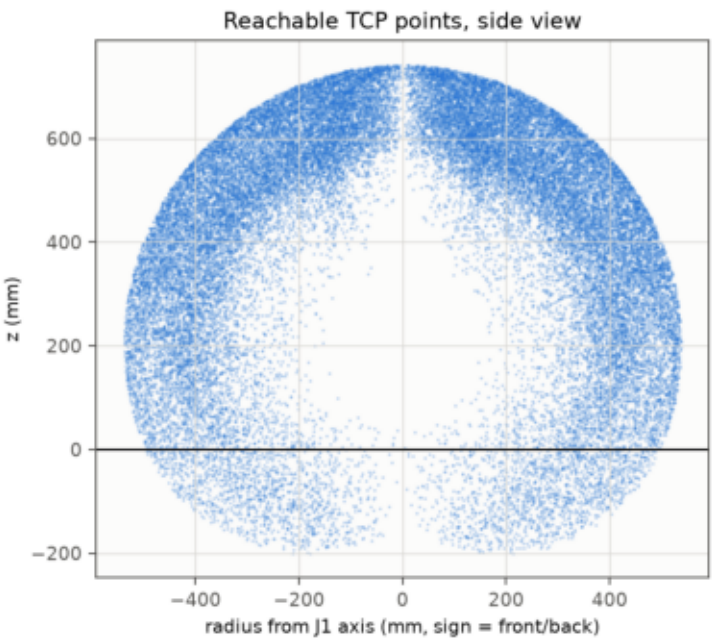
Kinematic model (design.py): each moving link has its frame on the proximal joint axis, with the joint axis as local Z. Forward kinematics chains the fixed parent-to-child frames with Rz(q). The same frames drive the Inventor assembly: every joint is an axis mate, a flush and an angle constraint whose value is an expression of the assembly parameters q1..q6.

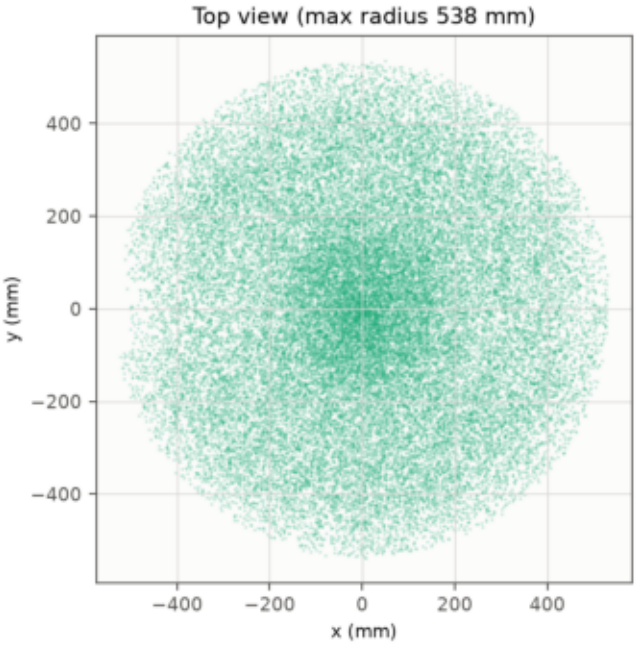
Validation in CAD: Inventor's solver was driven to 11 poses (home plus 10 random poses inside the limits). The TCP from the solved occurrences matches the Python FK to 0.00043 mm (link poses to 1e-3 mm). This is a measurement on the CAD model, not on hardware.

Workspace (40000 random samples in the joint limits): max TCP radius 538 mm, TCP z from -201 to 742 mm. With the tool pointing down at the 25 mm grasp height it reaches out to 390 mm from the base axis.

Joint limits used (deg, measured collision-free in Inventor where noted): q1 [-170, 170] *, q2 [-60, 110] *, q3 [-100, 100] *, q4 [-90, 90] *, q5 [-35, 100] *, q6 [-135, 135] * (* = Inventor interference sweep)

Workspace


Reachable TCP points, side view
600
400
200
0
−200
z (mm)
−400
−200
0
200
400
radius from J1 axis (mm, sign = front/back)


Top view (max radius 538 mm)
400
200
0
−200
−400
y (mm)
−400
−200
0
200
400
x (mm)

3 Mass model (from CAD)

Masses are Inventor mass properties: solid printed parts at the filament density (PA6-CF 1.09, PETG 1.27 g/cm3), with datasheet or estimated mass overrides on purchased parts (steppers 0.55/0.60 kg estimated, servos 60/65 g, bearings from the catalogue). Each occurrence is assigned to the most distal joint that moves its CG in Inventor.

Link group	Mass g	CG (link frame) mm	members
Base	1820	[-20.8, -0.2, 52.7]	15
Turntable	1291	[-31.2, 47.0, 56.3]	17
UpperArm	1058	[104.9, -0.0, -37.2]	19
Forearm	472	[76.3, -6.5, -3.3]	18
WristYoke	160	[-6.0, 0.4, 18.7]	13
WristBracket	107	[33.9, -6.8, 2.7]	7
Gripper	107	[-14.5, -5.2, 28.8]	10

Moving mass 1.90 kg; payload share of the J2 torque at full extension 40%.

4 Drive train: gears, belt, bearings, shafts

Spur gears: involute 20 deg teeth modelled in Inventor (spline flanks, circular pattern), with root relief below the base circle and 0.06 mm backlash. In the assembly each pinion is angle-constrained to its joint through the ratio and the mesh phase, so the teeth stay in mesh at any pose. Inventor interference checks at the test poses show no tooth overlap (see section 6). Lewis bending stress with the Barth velocity factor, at the design torque (J1 at its full capacity):

Pair	m	teeth	face	T joint	Ft N	Kv	s pin	s gear	SF pin
J1	1.5	18:72	16.0	7.09	131	1.006	17.8	12.7	1.68
J2	1.75	18:84	14.0	6.54	89	1.005	11.8	8.3	2.54
J4	1.25	16:24	8.0	0.84	56	1.0	18.9	16.6	1.59
J5	1.0	20:40	10.0	0.85	42	1.002	13.2	10.9	2.27

Allowable tooth stress 30.0 MPa (assumed, printed PA6-CF bending fatigue).

Belt HTD 375-5M-15: centre distance 109.91 mm, effective tension 78 N, recommended pretension 59 N/strand, tight side 138 N, SF 2.18 against an assumed 300 N working limit, 6.5 teeth in mesh on the 15T pulley (6 or more recommended).

Bearings: shoulder 6201 radial load 38 N (x1.5), static safety s0 = 80.8. loads are tens of N at <10 rpm: static safety s0 governs, life is not a concern.

Shoulder shaft dia 12 steel: M = 0.60 N.m, sigma = 3.5 MPa, SF 98.9 vs 350 MPa yield.

5 Motion simulation

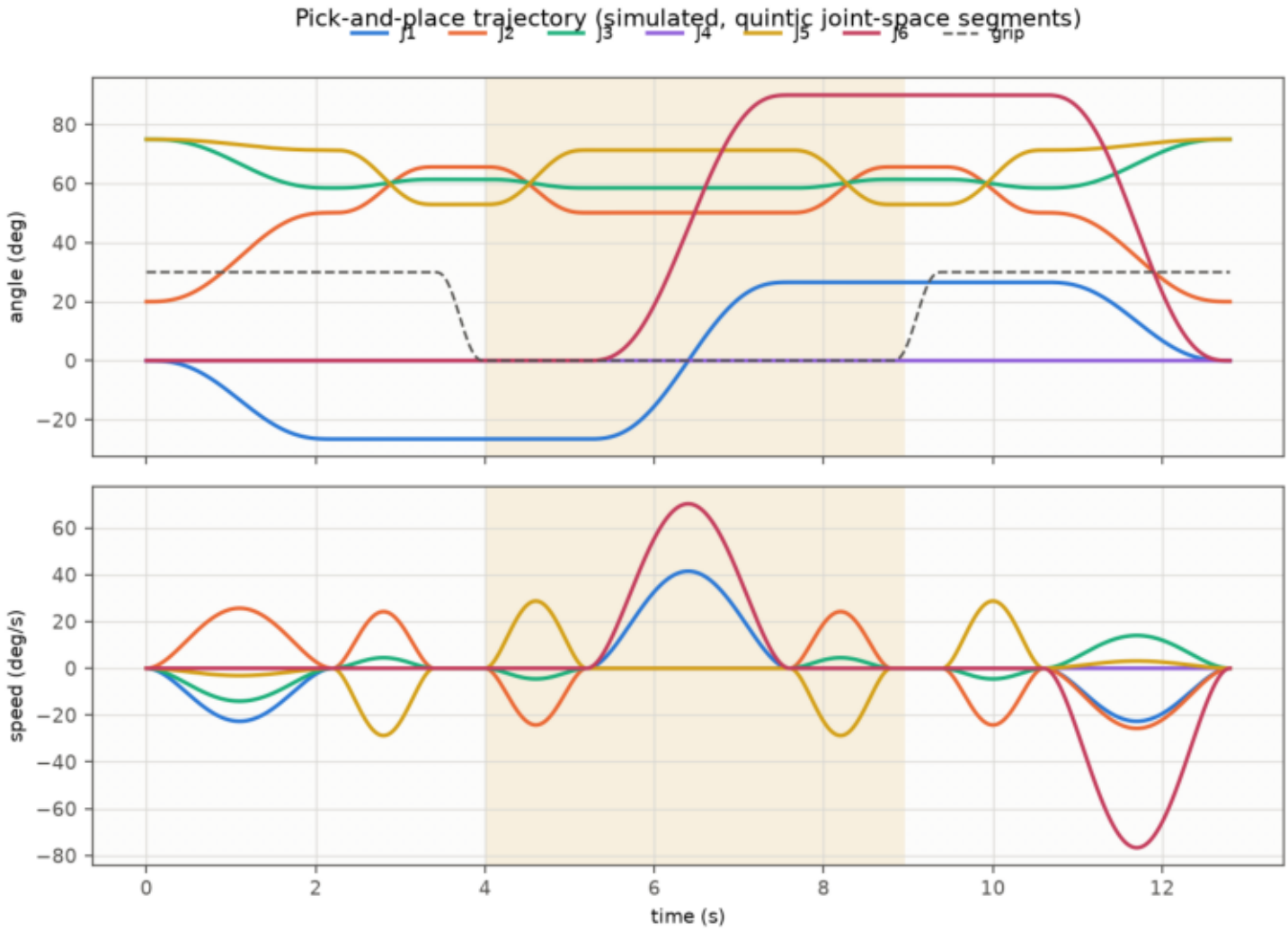
Pick-and-place cycle (simulated): home, then pick at [300.0, -150.0, 25.0], place at [300.0, 150.0, 25.0] mm, tool pointing down, 0.5 kg payload held from grasp to release, J6 rotated 90 deg during transfer, 12.8 s total. Waypoints come from closed-form IK (planar 2R plus wrist); quintic joint-space segments (zero velocity and acceleration at each waypoint). Peak TCP speed 0.24 m/s.

Dynamics: recursive Newton-Euler in the world frame with the six CAD link groups (mass, CG, inertia tensor from Inventor) and the payload as a point mass. Motor-side demand includes the rotor inertia reflected through the ratio.

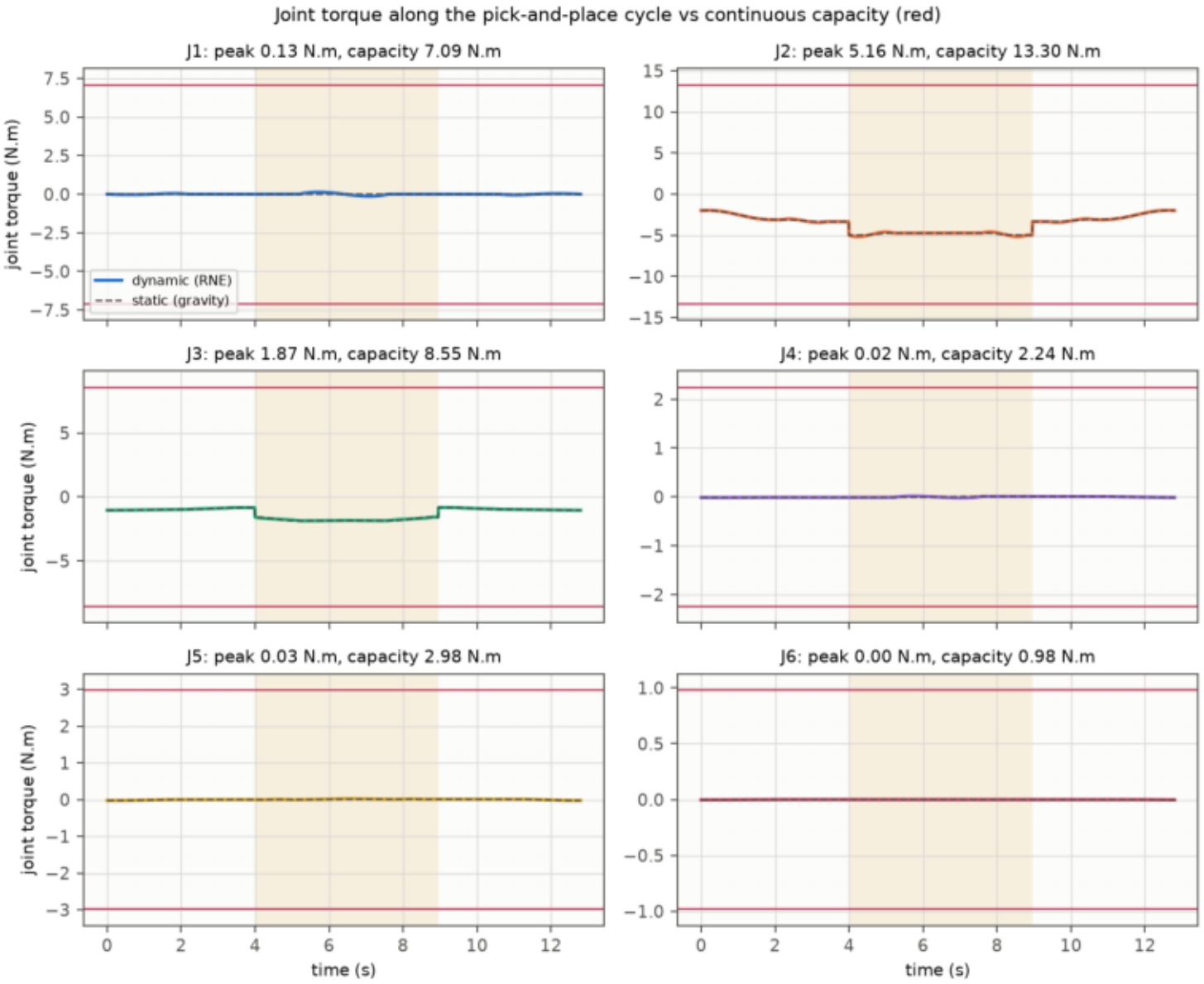
Dynamic/static torque ratio along the cycle: J1 n/a (no gravity load), J2 1.03, J3 1.00, J4 n/a (no gravity load), J5 n/a (no gravity load), J6 n/a (no gravity load). The FEA uses a dynamic factor of 1.5.

Inventor motion study: the assembly parameters were stepped through 18 frames of this trajectory and AnalyzeInterference was run on all occurrences at each frame: 0 interference hits (> 0.01 mm3) over the cycle.

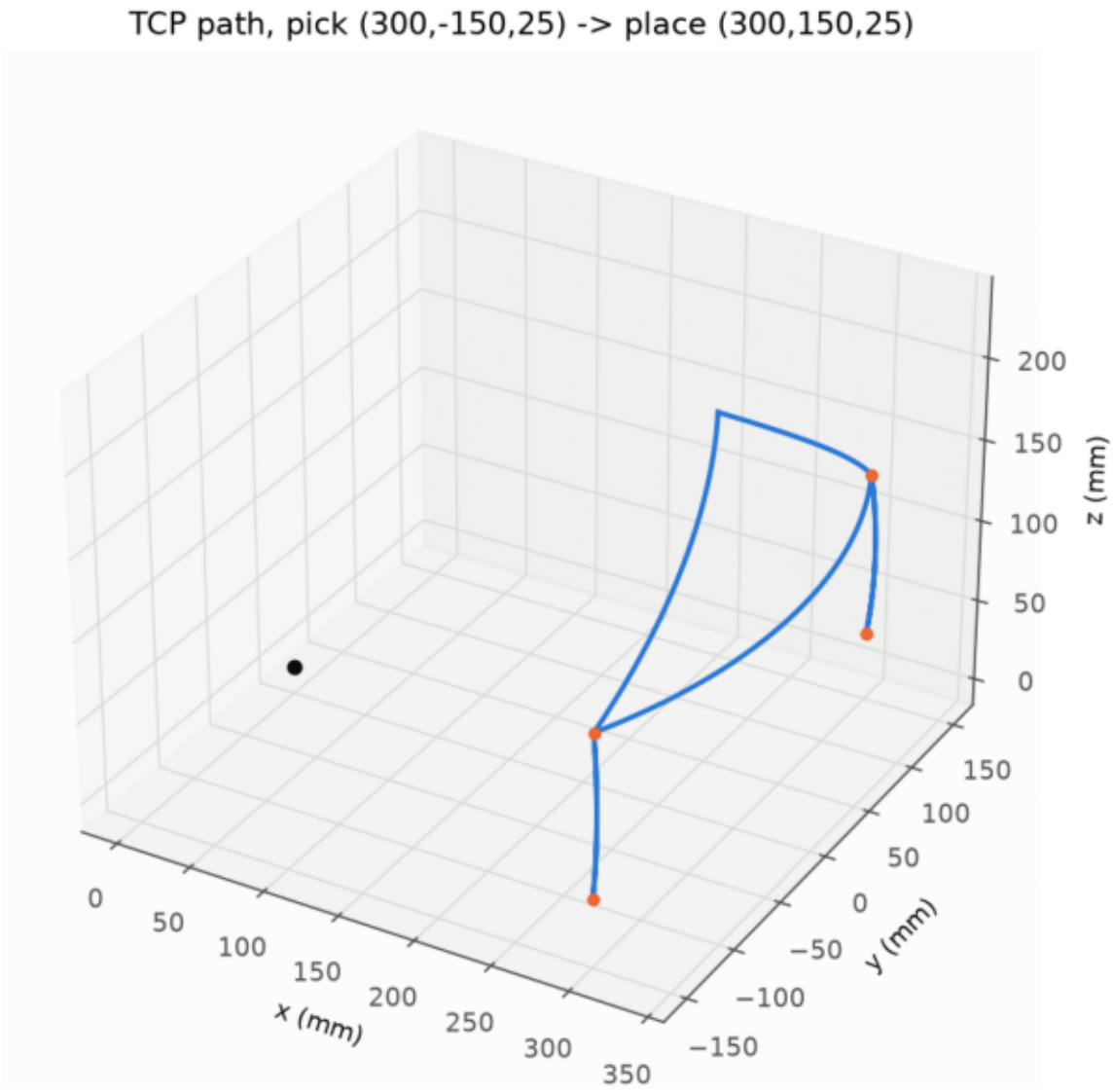
Joint angles and speeds



Joint torques vs capacity



TCP path


TCP path, pick (300,-150,25) -> place (300,150,25)
x (mm)
y (mm)
z (mm)
0
50
100
150
200
250
300
350
−150
−100
−50
0
50
100
150
200

Inventor motion study frames



6 Structural FEA (simulated)

Solver: gmsh (OpenCASCADE STEP import from the Inventor parts) + scikit-fem, quadratic P2 tetrahedra, linear elastic, PyPardiso direct solve (the same chain as QuadDrone). Material: Bambu PA6-CF, conditioned values E 3560 MPa, nu 0.35. The design allowable is 45 MPa (layer-direction bending, conditioned); XY tensile is 102 MPa (dry). Parts are analysed as solid bodies.

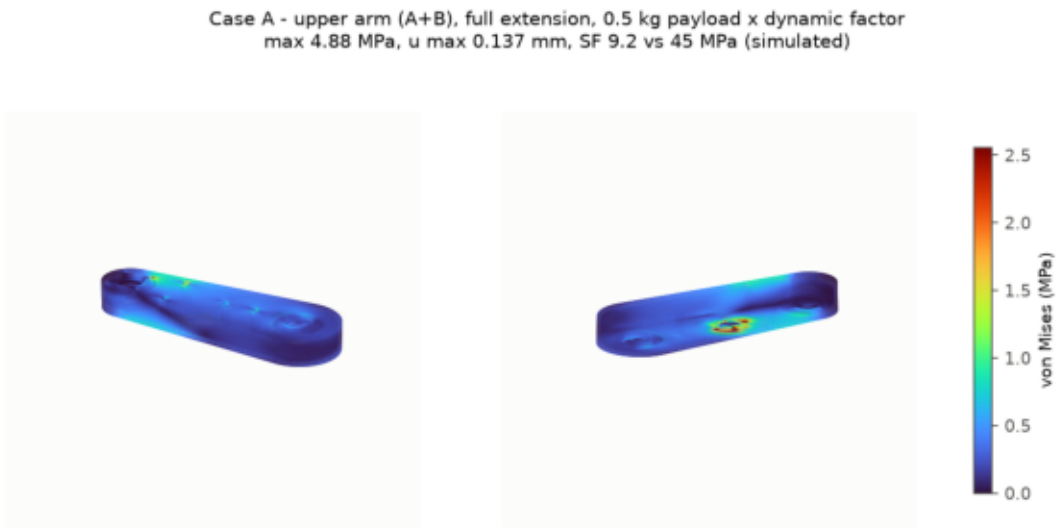
Load case: full horizontal extension with the 0.5 kg payload (largest J2 and J3 torque), forces from the Newton-Euler model times a dynamic factor of 1.5. Bearing loads are applied as uniform traction on the seats, and tilting moments as linear normal traction. Fixed boundary conditions are rigid, so the peak next to them is a stress concentration; the value away from the constraint zone is also reported.

Case	DOF	vm max	vm away	u max	SF 45	SF away	SF XY 102
caseA_upper_arm	643,200	4.88	4.88	0.137	9.2	9.2	20.9
caseB_shoulder_cheek	358,179	12.95	12.95	0.459	3.5	3.5	7.9
caseC_forearm	429,006	9.08	9.08	0.216	5.0	5.0	11.2

Case A elbow deflection 0.062 mm under load; belt pull 275 N.

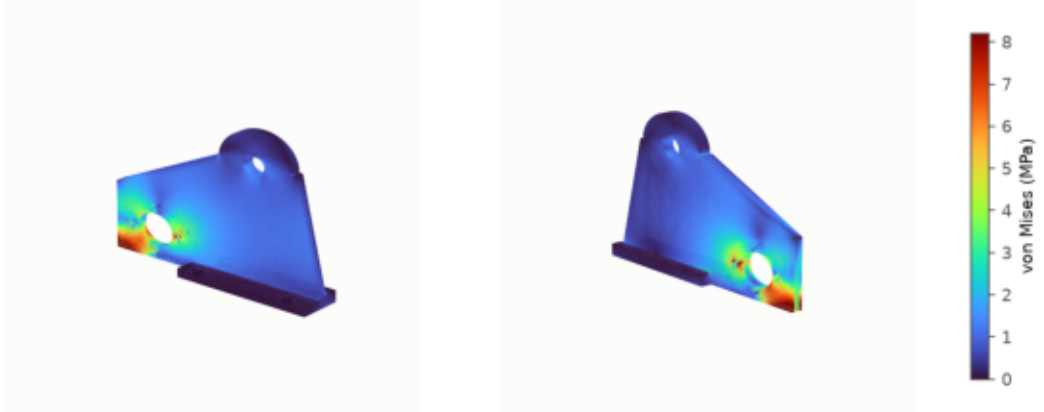
Case C nose deflection 0.146 mm; wrist moment [0.0, 0.0, 1.792] N.m on a single 6806 bearing. A second bearing or a 4-point-contact bearing is recommended for production.

FEA case A


Case A - upper arm (A+B), full extension, 0.5 kg payload x dynamic factor
max 4.88 MPa, u max 0.137 mm, SF 9.2 vs 45 MPa (simulated)
2.5
2.0
1.5
1.0
0.5
0.0
von Mises (MPa)

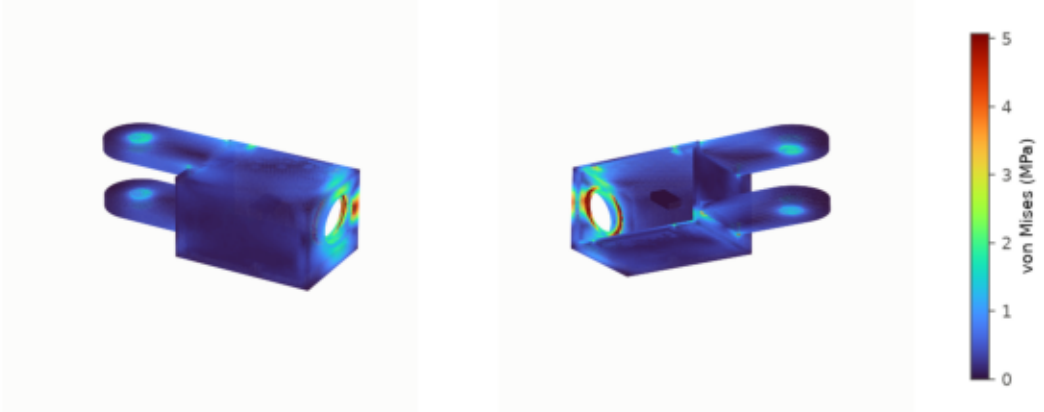
FEA case B

Case B - shoulder cheek (J2 motor side)
max 12.95 MPa, u max 0.459 mm, SF 3.5 vs 45 MPa (simulated)



FEA case C

Case C - forearm, wrist reaction on the 6806 seat
max 9.08 MPa, u max 0.216 mm, SF 5.0 vs 45 MPa (simulated)



7 Assembly checks (Inventor)

DOF: {'0T/0R': 99}; constraints 193 {'Flush': 109, 'Insert': 52, 'Mate': 19, 'Angle': 13}; unhealthy 0.

Interference at home: 4 pairs, 0.001 mm3 (residual = numerical tangency of M3 screws in nominal-diameter holes)

Interference at random1: 4 pairs, 0.001 mm3 (residual = numerical tangency of M3 screws in nominal-diameter holes)

Interference at random2: 4 pairs, 0.001 mm3 (residual = numerical tangency of M3 screws in nominal-diameter holes)

Interference at reach_horizontal: 4 pairs, 0.001 mm3 (residual = numerical tangency of M3 screws in nominal-diameter holes)

Interference at pick_low: 4 pairs, 0.001 mm3 (residual = numerical tangency of M3 screws in nominal-diameter holes)

Sweep q1 (others q2=20, q3=75, q4=0, q5=75, q6=0): collision-free [-170.0, 170.0] of [-170, 170]; limiting contacts:

Sweep q2 (others q1=0, q3=75, q4=0, q5=75, q6=0): collision-free [-60.0, 110.0] of [-60, 130]; limiting contacts: BaseHousing/BeltHTD5M_375, BaseHousing/UpperArmA, BaseHousing/UpperArmB

Sweep q2 (others q1=0, q3=0, q4=0, q5=0, q6=0): collision-free [-60.0, 110.0] of [-60, 130]; limiting contacts: BaseHousing/BeltHTD5M_375, BaseHousing/UpperArmA, BaseHousing/UpperArmB

Sweep q3 (others q1=0, q2=20, q4=0, q5=75, q6=0): collision-free [-100.0, 100.0] of [-150, 150]; limiting contacts: FingerL/ShoulderCheekL, FingerL/Turntable, FingerR/J2Gear84T

Sweep q3 (others q1=0, q2=90, q4=0, q5=0, q6=0): collision-free [-100.0, 100.0] of [-150, 150]; limiting contacts: Bearing608_2RS/BaseHousing, Bearing608_2RS/ScrewM4x12, Bearing608_2RS/Turntable

Sweep q4 (others q1=0, q2=20, q3=75, q5=75, q6=0): collision-free [-180.0, 180.0] of [-180, 180]; limiting contacts:

Sweep q5 (others q1=0, q2=20, q3=75, q4=0, q6=0): collision-free [-60.0, 100.0] of [-130, 130]; limiting contacts: Bearing6806_2RS/ScrewM3x10, Bearing6806_2RS/ServoDS3218, Bearing6806_2RS/WristBracket

Sweep q5 (others q1=0, q2=20, q3=75, q4=90, q6=0): collision-free [-60.0, 100.0] of [-130, 130]; limiting contacts: Bearing6806_2RS/ScrewM3x10, Bearing6806_2RS/ServoDS3218, Bearing6806_2RS/WristBracket

Sweep q6 (others q1=0, q2=20, q3=75, q4=0, q5=75): collision-free [-180.0, 180.0] of [-180, 180]; limiting contacts:

Sweep grip (others q1=0, q2=20, q3=75, q4=0, q5=75, q6=0): collision-free [0.0, 45.0] of [0, 45]; limiting contacts:

8 Design for 3D printing

Process: FDM on a Bambu Lab printer, 256 x 256 mm bed, 0.4 mm nozzle. PA6-CF for structural parts and gears, PETG for the base and the cover. One plain 3MF per plate sits next to the assembly (Jay slices each in Bambu Studio).

BasePlate: flat; watertight True; wall min/p05 2.5/6.0 mm (pass); overhang <45 deg 0.1% (no supports)

BaseHousing: deck down; watertight True; wall min/p05 0.568/3.977 mm (pass); overhang <45 deg 0.6% (no supports)

Turntable: disc face down, teeth up; watertight True; wall min/p05 1.791/2.545 mm (pass); overhang <45 deg 0.1% (no supports)

ShoulderCheekR: foot down, plate vertical; watertight True; wall min/p05 1.814/6.0 mm (pass); overhang <45 deg 1.0% (no supports)

ShoulderCheekL: foot down, plate vertical; watertight True; wall min/p05 4.761/6.0 mm (pass); overhang <45 deg 0.3% (no supports)

UpperArmA: outer wall down; watertight True; wall min/p05 1.0/2.746 mm (pass); overhang <45 deg 1.7% (supports needed)

UpperArmB: outer wall down; watertight True; wall min/p05 1.0/2.197 mm (pass); overhang <45 deg 1.3% (supports needed)

Forearm: -z side down; tree supports under the +z fork plate; watertight True; wall min/p05 0.523/4.0 mm (pass); overhang <45 deg 17.0% (supports needed)

ForearmCover: flat; watertight True; wall min/p05 0.303/3.0 mm (pass); overhang <45 deg 0.0% (no supports)

WristYoke: hub down, tree supports under the base plate; watertight True; wall min/p05 2.295/4.0 mm (pass); overhang <45 deg 20.6% (supports needed)

WristBracket: crossbar down, arms vertical; watertight True; wall min/p05 1.175/1.751 mm (pass); overhang <45 deg 4.8% (supports needed)

GripperBase: flange down, plate vertical; watertight True; wall min/p05 1.405/4.25 mm (pass); overhang <45 deg 3.7% (supports needed)

FingerL: flat; watertight True; wall min/p05 1.087/1.412 mm (pass); overhang <45 deg 0.0% (no supports)

FingerR: flat; watertight True; wall min/p05 1.083/1.496 mm (pass); overhang <45 deg 0.0% (no supports)

J2Gear84T: outer face down; watertight True; wall min/p05 2.098/2.452 mm (pass); overhang <45 deg 0.3% (no supports)

J1Pinion18T: flat; watertight True; wall min/p05 1.691/1.848 mm (pass); overhang <45 deg 0.0% (no supports)

J2Pinion18T: flat; watertight True; wall min/p05 1.988/2.361 mm (pass); overhang <45 deg 0.0% (no supports)

J4Pinion: flat; watertight True; wall min/p05 1.402/1.505 mm (pass); overhang <45 deg 0.0% (no supports)

J5Pinion: flat; watertight True; wall min/p05 1.093/1.321 mm (pass); overhang <45 deg 0.0% (no supports)

J4Gear: flat; watertight True; wall min/p05 0.908/1.529 mm (pass); overhang <45 deg 1.7% (supports needed)

PulleyHTD5M_45T: flat; watertight True; wall min/p05 0.147/1.823 mm (pass); overhang <45 deg 0.0% (no supports)

plate 1 [PETG]: 240 x 240 x 6 mm, 1 parts: BasePlate (flat)

plate 2 [PETG]: 200 x 200 x 90 mm, 1 parts: BaseHousing (deck down)

plate 3 [PETG]: 95 x 74 x 3 mm, 1 parts: ForearmCover (flat)

plate 4 [PA6-CF]: 159 x 255 x 20 mm, 1 parts: UpperArmA (outer wall down)

plate 5 [PA6-CF]: 255 x 159 x 20 mm, 1 parts: UpperArmB (outer wall down)

plate 6 [PA6-CF]: 242 x 178 x 105 mm, 3 parts: Turntable (disc face down, teeth up), ShoulderCheekR (foot down, plate vertical), ShoulderCheekL (foot down, plate vertical)

plate 7 [PA6-CF]: 171 x 224 x 74 mm, 2 parts: J2Gear84T (outer face down), Forearm (-z side down; tree supports under the +z fork plate)

plate 8 [PA6-CF]: 254 x 140 x 80 mm, 11 parts: WristYoke (hub down, tree supports under the base plate), PulleyHTD5M_45T (flat), WristBracket (crossbar down, arms vertical), GripperBase (flange down, plate vertical), FingerL (flat), FingerR (flat), J2Pinion18T (flat), J4Gear (flat), J1Pinion18T (flat), J4Pinion (flat), J5Pinion (flat)

9 Assumptions and limitations

Payload 0.5 kg treated as a point mass at the TCP (between the finger pads); 20 mm CG offset assumed for J4/J6 checks.

Printed parts modelled solid at the filament density (PA6-CF 1.09, PETG 1.27 g/cm3): conservative (heavier) for thin-wall parts.

Stepper pull-out curve ASSUMED (typical 17HS19-class at 24 V); holding torque 0.52 N.m and gearbox ratings (3 N.m max permissible, 5 N.m moment) from StepperOnline listings.

Servo stall torques from DSSERVO listings at 6 V; continuous design budget = 50 % of stall.

Efficiencies: PG5 90 %, PG19 80 % (datasheet); printed spur stage 95 %, HTD belt 95 % (assumed).

Motor and gearbox masses: 0.55 kg (PG5) / 0.60 kg (PG19) estimated; servo masses from listings.

Printed gear tooth allowable 30 MPa (PA6-CF, bending fatigue, assumed ~0.3 x XY UTS); Lewis form factor, Barth velocity factor.

HTD 5M 15 mm allowable working tension taken as 300 N (conservative assumption; check the belt supplier's table).

Servos are modelled from datasheet envelopes (40 x 20 x 40.5 mm case, 54.5 mm ears, spline 10 mm off-centre). The stepper body is the step.parts NEMA17 48 mm model; gearbox geometry and the 4x M3 front pattern (PCD 30) are modelled from typical dimensions and must be checked against the vendor drawing.

The vendor 15T pulley (step.parts) is a simplified cylinder without teeth; belt teeth are not modelled.

FEA: linear elastic, isotropic, solid parts; printed anisotropy is covered only through the 45 MPa Z-direction allowable. Bolted joints are idealised as bonded or fixed.

Nothing in this report is a physical test. Joint torques, speeds, stresses and deflections are model predictions; the first prototype should verify belt tension, gear backlash and J2 holding with the payload.